

TOPIC 01: WHAT IS GIT & GITHUB?

What is Git?

Git ek **Free, Open-Source Version Control System (VCS)** hai jo hamare codebase ke changes ko track karta hai aur project history maintain karta hai.

What is GitHub?

GitHub ek online cloud platform/website hai jahan hum apne Git repositories ko host karte hain aur doosre developers ke sath collaborate karte hain.

GIT (LOCAL TOOL ON PC)

→ Push to Web →

GITHUB (CLOUD HOST / WEBSITE)

TOPIC 02: FILE STATUS LIFECYCLE

Git Ke 4 File States:

UNTRACKED (NAYI FILE)

→ git add →

STAGED (READY FOR COMMIT)

↓ git commit

UNMODIFIED (SAVED IN REPO)

→ Edit file →

MODIFIED (CHANGES MADE)

Core 3-Step Routine:

```
$ git add index.html # Step 1: Working directory se Staging area me bhejo
$ git commit -m "save msg" # Step 2: Staging se Local Repository me commit karo
$ git push origin main # Step 3: Local Repo se GitHub (Remote) par push karo
```



Key Rule: git add . saare modified & untracked files ko ek sath stage kar deta hai!

TOPIC 03: BASIC COMMANDS (CLONE & STATUS)

- `git clone <repo_url>` → Remote repository ko download karke local machine par copy banata hai.
- `git status` → Code ki current state display karta hai (kaunsi file modified hai, staged hai ya untracked hai).

TOPIC 04 & 05: NEW REPO & REMOTE SETUP

First Time Repo Setup Steps:

```
$ git init # Naya empty Git repo initialize karta hai $ git remote add origin <repo_url> # Local repo ko remote GitHub link se jodta hai $ git remote -v # Remote URL verify/check karta hai $ git branch # Saari branches check karta hai $ git branch -M main # Current branch ka naam rename karke "main" karta hai $ git push -u origin main # Pehli baar code remote 'origin' par push karta hai
```

TOPIC 06: BRANCH COMMANDS

Branching Kya Hai? Project ki main line se alag hokar safe features banane ka isolated rasta.

Command	Kaam (Action)
git branch	Active branches list karta hai (* current branch par hota hai).
git checkout <branch>	Dusri branch par switch/navigate karta hai.
git checkout -b <new_branch>	Nayi branch banata hai AUR turant uspar switch karta hai.
git branch -d <branch>	Purani/merged branch ko delete karta hai.



Modern Git Tip: Modern Git me git switch <branch> aur git switch -c <new_branch> bhi use hota hai!

TOPIC 07: MERGING CODE & DIFF

- git diff <branch> → Do branches, commits ya working tree ke beech changes ko compare karta hai.
- git merge <branch_name> → Specify ki gayi branch ke code ko current branch ke sath combine/merge karta hai.

```
$ git checkout main # 1. Main branch pe aao $ git merge feature-login # 2. Feature branch ko main me merge karo
```

TOPIC 08: GIT PULL ORIGIN MAIN

Pull = Fetch + Merge

Remote repository se latest code download (fetch) karta hai aur turant current local branch ko update (merge) kar deta hai.

```
$ git pull origin main # Team members ke naye commits apne paas lana
```

TOPIC 09: FIXING MISTAKES (UNDOING IN GIT)

Case	Problem	Solution Command
Case 1	Galti se file stage ho gayi (git add ke baad unstage karna)	git reset <filename> ya git restore --staged <file>
Case 2	Aakhri 1 commit galat ho gaya (Commit wapas lena par code rakhna)	git reset HEAD~1 (Soft: changes rehte hain, commit hat jata hai)
Case 3	Multiple commits purane kisi specific point par wapas jana	git reset <commit_hash>



Caution: git reset --hard saare uncommitted changes ko permanently mita deta hai, dhyan se chalayein!

TOPIC 10: WHAT IS FORKING?

Fork Concept:

Fork kisi doosre developer ke public GitHub repository ki ek independent **Personal Copy** hoti hai jo aapke apne account me save ho jaati hai.

ORIGINAL UPSTREAM REPO

→ Fork →

YOUR ACCOUNT COPY (ROUGH COPY)

↓ (Changes karke)

SEND PULL REQUEST (PR) TO ORIGINAL

- Yeh original repo ke code aur visibility ko inherit karti hai.
- Open source me contribute karne ke liye pehle Fork kiya jata hai, phir branch bana kar changes karte hain.

